

Today's data is on house pricing in Los Angeles! We have access to data on all of the houses that sold in four cities around west LA in the course of a single month. Our goal is to predict the price of a home based on a few attributes of that home. The LA data frame is available in the Files section of bCourses.

1. Create a scatterplot examining the relationship between square footage and price.
 - Write the code you used below.

 - Sketch the plot below.

2. Create a scatter plot examining the relationship between square footage *squared* (raised to the second power). Sketch the plot below.

3. Add two columns onto the LA dataset, `log_price` and `log_sqft`, which measure *log* price and *log* square footage, respectively. Use the *natural log*. Save the new dataset back into itself.

4. Create another scatterplot examining the relationship between *log* square footage and *log* price.
 - Sketch the plot below.

5. Rather than writing the code you used, explain in at least two sentences whether the logged variables or the regular variables appear more suitable for a linear model.

6. Fit three linear models

- one which predicts price with square footage;
- one which predicts price with the square footage raised to the second power (square footage squared);
- one which predicts log price with log square footage.

You do not need to copy the code below, but, using the broom library, find and report the R^2 for both models below.

Model	R^2
price ~ sqft	
price ~ sqft^2	
log_price ~ log_sqft	

7. In one to two sentences, explain whether the results you have line up with your explanation in **Question 5**.

Recently the University of California purchased a new house to serve as the residence of the university President and to host university functions. The address of the house is 2821 Claremont Blvd in Berkeley. You can find a listing for the house on the Internet!

8. Use the linear model with the highest R^2 from above to predict the sale price in (USD) of this house.
Hint: depending on which model you select, you may have to do another step after getting an initial output.

9. Was your model an under- or over-estimate? Why do you think this is? Answer in at least two sentences.